

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Enhanced Uni-Directional Time Sync changes Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Adding a recommendation for a new Time-Offset Filter

The addition of a new time-offset filter will eliminate the need for multiple resolution levels (HiFi/MedRes) and allow the use of a single parameter set in Enhanced Uni-directional Time-Sync. The single set will reduce the minimum time in CL0 requirement and allow for 50ms of CLx. Since the known use cases do not require 1 ns accuracy the allowed noise was enlarged to 2 ns.

Benefits as a result of the changes:

Simplifying TMU modes complexity, CM configurations, and router validation and certification, while enabling CLx entry for up to 50ms.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

NA

An analysis of the hardware implications:

NA

An analysis of the software implications:

NA

An analysis of the compliance testing implications:

Remove tests with MedRes configuration

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Actual Change

(a). Section 7.3.1 Time Sync Handshake

To Text:

Enhanced Uni-Directional Time Sync Handshakes use ~~either HiFi Mode or MedRes Mode~~. The mode is configured using the *AdapterTimeSyncInterval*, *ReplenishN*, *ReplenishTimeout*, and *ReplenishThreshold*; ~~and~~ *DeltaAvgConst* fields.

A Router shall support Bi-Directional Time Sync Handshakes in HiFi Mode. A Router shall support Uni-Directional Time Sync Handshakes in HiFi Mode and LowRes Mode. A Router shall support Enhanced Uni-Directional Time Sync Handshake in ~~both HiFi Mode and MedRes Mode~~.

(b). Section 7.4.1 Intra-Domain Equations

To Text:

- ~~An Upstream Facing Port uses Equation 7-11 (Time Offset Delta) to calculate the current change in Time Offset between two consecutive handshakes.~~
- ~~An Upstream Facing Port uses the Time Offset Delta in Equation 7-12 (First Time Offset from Downstream Facing Port) on handshakes that SleepCyclesN field in the Follow-Up Packet does not equal 0. This calculation shall not be filtered.~~

...

~~Equation 7-11. Time Offset Delta from DFP for Two Consecutive Calculations~~

$$\Delta O(U_i, D_i) = O(U_i, D_i)[n] - O(U_i, D_i)[n-1]$$

~~Equation 7-12. First Time Offset from DFP after a Pause~~

$$O(U_i, D_i)[n] = O(U_i, D_i)[n - \text{SleepCyclesN} - 1] + (\text{SleepCyclesN} + 1) * \Delta O(U_i, D_i)$$

~~Note: A Time Offset Delta value is required in order to calculate the first time offset after a pause. The Time Offset Delta value is computed from the time offset difference between the n and (n-1) handshakes. During a pause, TMU handshakes do not take place, so the Delta and Time Offset Delta calculations are not done by the Upstream Facing Port. The Upstream Facing Port calculates the Delta and Time Offset Delta values for the TMU handshakes before and after the pause, but not during the pause. In Equation 7-11, $\Delta O(U_i, D_i)$ is the Delta that was calculated by the Upstream Facing Port for the last handshake before the pause.~~

(c). Section 7.4.3 Filtering

To Text:

In order to achieve the required accuracy, low pass it is recommended to use the following filters in the process of computing: ~~are applied on some of the computations.~~ The parameters that need to be

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~~filtered are~~ Propagation Delay (Tpd), Time Offsets, ~~Time Offset Delta, and~~ Frequency Offset computations, ~~and Time Offset Delta~~. The recommended filter is an IIR filter as follows:

Section 7.4.3.1 describes the first order IIR filter that should be applied on Propagation Delay (Tpd), and Frequency Offset computations. Section 7.4.3.2 describes the second order IIR that should be applied on Time Offset computation.

7.4.3.1 First order IIR

The first order IIR filter is described in Equation 7-22:

Equation 7-22. ~~1st~~First order IIR Low Pass Filter

$$Y[n] = \left(1 - \frac{1}{2^p}\right) \cdot Y[n-1] + \frac{1}{2^p} \cdot X[n]$$

where:

- $X[n]$ is the new input to filter.
- $Y[n-1]$ is the previous value of the filter.
- $Y[n]$ is the new value of the filter.
- p is the filter strength.

The values in the following registers in Router Configuration Space can be used for filtering frequency offset, and propagation delay:

- FreqAvgConst.
- DelayAvgConst.

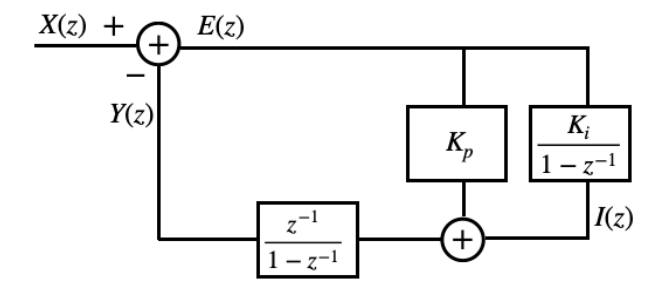
It is recommended that a USB4 Port apply IIR filters with strength (p) of at least 4.

7.4.3.2 Second order IIR filter

Time Offset filtering should utilize a filter capable of autonomously estimating noise without compromising Time Offset accuracy. The second order IIR filter described below will effectively learn the noise characteristics, eliminating noise interference after an initial convergence phase without attenuating the Time Offset.

Figure X-1 illustrates the filter, and equations W-1, W-2, and W-3 describe the steps to calculate the filter output.

Figure X-1. 2nd order IIR Filter



Equation W-1

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$$E[n] = X[n] - Y[n]$$

Equation W-2

$$I[n] = I[n - 1] + E[n] \cdot K_i$$

Equation W-3

$$Y[n + 1] = Y[n] + E[n] \cdot K_p + I[n]$$

where:

- X[n] is the new input to filter.
- Y[n] is the new value of the filter.
- Y[n+1] is the next value of the filter in the following handshake.
- $K_p \triangleq 2^{-P_p}$ and $K_i \triangleq 2^{-P_i}$ while p_p and p_i are the filter strengths.

7.4.3.2.1 Calculations on handshake resumption

During As result of handshake suspension (for example during CLx residency, or missing handshakes due to link errors), the filter is not being updated. To compensate for this period and calculate the new value of the filter, the following estimation is used by using equations Z1 and Z2. This estimation is applied only in the first handshake after pause.

Equation Z-1

$$I[n - 1] = I[n - SleepCyclesN - 1]$$

Equation Z-2

$$Y[n] = Y[n - SleepCyclesN] + SleepCyclesN \cdot I[n - 1]$$

where:

- SleepCycleN is the number of skipped handshakes(s).
- n-SleepCycleN-1 is the last handshake index before handshake(s) was skipped.
- I[n-SleepCycleN-1] was calculated in handshake [n-SleepCycleN-1].
- Y[n-SleepCycleN] was calculated in handshake [n-SleepCycleN-1].

Subsequently, to complete the compensation calculations (in the first handshake after pause), equations W-1, W-2, and W-3 should be used to compute Y[n+1] and I[n]. In the following handshakes the filter returns to its normal operation using only equations W-1, W-2, and W-3

7.4.3.2.2 Filter's strength

The values in the following registers in Router Configuration Space can be used for filtering frequency offset, and propagation delay:-

- OffsetPpAvgConst.
- OffsetPiAvgConst.

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For USB4 Ports, employing IIR filters with $P_p = 5$ and $P_i = 17$ is recommended. However, in instances of missed handshakes or power-up events, adaptive behavior is advised. Table W below outlines the Filter's strength from the initial handshake until the convergence period's conclusion.

Figure W converge phase strengths

Handshake Index	P_p	P_i
1	Filter invalid	Filter invalid
2-3	1	6
4-7	2	7
8-15	3	8
16-31	4	9
32-63	5	10
64-127	5	11
128-255	5	12
256-511	5	13
512-1023	5	14
1024-2047	5	15
2048-3125	5	16
3126 and afterwards	5	17

~~In order to compensate for filter attenuation, the error value is computed as follows and passed through IIR low pass filter:~~

$$E[n] = X[n] - Y[n]$$

$$E_{\text{filtered}}[n] = \text{IIR}(E[n])$$

~~The filtered error is then added back to the filtered time stamp value.~~

~~The values in the following registers in Router Configuration Space can be used for filtering time stamps, error, frequency offset, and propagation delay:~~

- ~~• FreqAvgConst.~~
- ~~• DelayAvgConst.~~
- ~~• OffsetAvgConst.~~
- ~~• ErrorAvgConst.~~
- ~~• DeltaAvgConst.~~

~~It is recommended that a USB4 Port apply IIR filters with strength (p) of at least 4.~~

(d). Section 7.5 Time Synchronization Accuracy Requirements

To Text:

Time synchronization accuracy is measured in two ways: Paired Measurement and Standalone Measurement. Section 7.5.1 describes Paired Measurement. Section 7.5.2 describes Standalone Measurement.

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When a Router is configured for HiFi Mode Time Sync Handshakes, the time synchronization accuracy is tHiAccuracy. ~~When a Router is configured for MedRes Mode Time Sync Handshakes, the time synchronization accuracy is tMedAccuracy.~~ When a Router is configured for LowRes Mode Time Sync Handshakes, the time synchronization accuracy is tLowAccuracy.

A Router shall reach the required time synchronization accuracy within tConvergeTime after Time Sync Handshakes are enabled.

If Time Sync Handshakes are paused, an Upstream Facing Port shall reach the required time synchronization accuracy within tCLxConvergeTime after receiving a Delay Response Packet.

(e). Section 7.5.1 Paired Measurement

To Text:

In HiFi Mode, the Dynamic Noise between two Routers connected to one another shall not add more than tHiAccuracy. ~~In MedRes Mode, the Dynamic Noise between two Routers connected to one another shall not add more than tMedAccuracy.~~ In LowRes Mode, the Dynamic Noise between two Routers connected to one another shall not add more than tLowAccuracy.

(f). Section 7.5.2 Standalone Measurement

To Text:

In HiFi Mode, the noise between measurement points and the ideal line shall not be more than tHiAccuracy. ~~In MedRes Mode, the noise between measurement points and the ideal line shall not be more than tMedAccuracy.~~ In LowRes Mode, the noise between measurement points and the ideal line shall not be more than tLowAccuracy.

Note: This measurement is also used to analyze the frequency response of the noise during compliance testing.

In order to achieve the goal of ~~tHiAccuracy~~ 4 ns static noise, the budget for TxTimeToWire uncertainty shall not be more than 6.4 ns dynamic noise with no more than 6.4 ns static offset that changes after each power up. RxTimeToWire shall not be more than 12.8 ns dynamic noise with no more than 6.4 ns static offset that changes after each power up.

(g). Table 7-12. Time Synchronization Accuracy Parameters

To Text:

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Parameter	Description	Min	Max	Units
tHiAccuracy	Time synchronization accuracy in HiFi Mode.	--	±2	ns
tMedAccuracy	Time synchronization accuracy in MedRes Mode.	--	2	ns
tLowAccuracy	Time synchronization accuracy in LowRes Mode.	--	4	ns
tConvergeTime	Time, after Time Sync Handshakes are enabled, to reach the required time synchronization accuracy.	--	50	ms
tCLxConvergeTime	Time, after Time Sync Handshakes are resumed after pause, to reach the required time synchronization accuracy.	--	100	μs

(h). Figure Error! No text of specified style in document.-1. Structure of the TMU Router Configuration Capability

To Text:

TSInterDomainInterval	ErrorAvgConst	OffsetPiAvgConst OffsetAvgConst	DelayAvgConst	FreqAvgConst	1 5
InterDomain Computation Timestamp 31:0					1 6
InterDomain Computation Timestamp 63:32					1 7
I DNS	Reserved	OffsetPiAvgConst Reserved	InterDomain Computation Timestamp 79:64		1 8

(i). Table Error! No text of specified style in document.-4. TMU Router Configuration Capability Fields

To Text:

15	TMU_RTR_CS_15	5:0	FreqAvgConst This field contains the IIR filter co-efficient that shall be used to average the frequency ratio. The IIR filter co-efficient is given by $(1/2^{\text{FreqAvgConst}})$.	R/W	8
		11:6	DelayAvgConst This field contains the IIR filter co-efficient that shall be used to average the propagation delay. The IIR filter co-efficient is given by $(1/2^{\text{DelayAvgConst}})$.	R/W	8
		17:12	<u>OffsetAvgConst</u> <u>This field contains the IIR filter co-efficient that shall be used to average the time offset. The IIR filter co-efficient is given by $(1/2^{\text{OffsetAvgConst}})$.</u> OffsetPiAvgConst This field contains the IIR filter co-efficient Pi that shall be used to average the time offset.	R/W	8

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		23:18	ErrorAvgConst This field contains the IIR filter co-efficient that shall be used to average the time offset averaging error. The IIR filter co-efficient is given by $(1/2^{\text{ErrorAvgConst}})$.	R/W	8
		31:24	TSInterDomainInterval This field controls the time interval between transmitting Inter-Domain Time Stamp Packets. The time interval is given in microseconds and is equal to $(\text{TSInterDomainInterval} + 1) * \text{TSPacketInterval}$. <i>Note: TSPacketInterval is used regardless of the TMU mode.</i>	R/W	0
16	TMU_RTR_CS_16	31:0	Inter-Domain Time Stamp Low For an Inter-Domain Time Initiator: This field shall contain the least significant 32 bits of the computed value of the Inter-Domain time stamp. The time stamp shall be computed as described in Section Error! Reference source not found. For all other Routers: This field shall contain the least significant 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
17	TMU_RTR_CS_17	31:0	Inter-Domain Time Stamp Middle For an Inter-Domain Time Initiator: This field shall contain the middle 32 bits of the computed value of the Inter-Domain time stamp. The time stamp shall be computed as described in Section Error! Reference source not found. For all other Routers: This field shall contain the middle 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
18	TMU_RTR_CS_18	15:0	Inter-Domain Time Stamp High For an Inter-Domain Time Initiator: This field shall contain the most significant 32 bits of the computed value of the Inter-Domain time stamp. The time stamp shall be computed as described in Section Error! Reference source not found. For all other Routers: This field shall contain the most significant 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
		23:16	OffsetPpAvgConst This field contains the IIR filter co-efficient Pp that shall be used to average the time offset. Reserved	R/W	0
		30:24	Reserved	Rsvd	0
		31	Inter-Domain Not Supported (IDNS) This field shall indicate whether a Router supports Inter-Domain Time Synchronization: 0 – Supported 1 – Not Supported A Host Router may set this field to either 1 or 0. A Device Router shall set this field to 0.	RO	Host Router: Vendor Defined Device Router: 0

(j). Table Error! No text of specified style in document.-2. TMU Adapter Configuration Capability Fields

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To Text:

9	TMU_ADP_CS_9	7:0	ReplenishN This field contains the minimum number of successful handshakes that shall be completed after missing <i>Replenish Threshold</i> handshakes before removing the objection to CL1 state.	R/W	0
		15:8	DirSwitchN This field contains the number of handshakes that are required in Inversed Bi-Directional mode before the transition to Adaptive Uni-Directional mode. This field shall be greater than 1 when Enhanced Uni-Directional Time Sync Handshakes are enabled.	R/W	0
		31:16	AdapterTimeSyncInterval This field contains the time interval between successive Time Sync handshakes. The time interval is specified in units of 1 μ s with 100ppm accuracy. This field is only used when Enhanced Uni-Directional mode is enabled. There are two allowed values for this register: 0: Shall disable the initiation of Time Sync Handshake from the USB4 Port. 16: “HiFi” Mode or MedRes Mode (16 μ s). All other values are reserved.	R/W	0

(k). Section 8.2.2.2 TMU Adapter Configuration Capability

To Text:



CONNECTION MANAGER NOTE

A Ver. 2 Connection Manager shall set the values in the TMU Adapter Configuration Capability of an Upstream Facing Port before it sets the Configuration Valid bit of the Router to 1b.

A Connection Manager shall use the following values to configure the TMU to HiFi Accuracy when using Enhanced Uni-Directional mode:

- *Replenish Timeout = ~~1875~~3125*
- *Replenish Threshold = 0*
- *ReplenishN = ~~222~~30*
- *DirSwitchN = 255*
- *AdapterTimeSyncInterval = 16*
- ~~*OffsetAvgConst = 8*~~
- ~~*DeltaAvgConst = 8*~~

~~*A Connection Manager uses the following values to configure the TMU to MedRes Accuracy when using Enhanced Uni-Directional mode:*~~

- ~~• *Replenish Timeout = 3125*~~
- ~~• *Replenish Threshold = 25*~~
- ~~• *ReplenishN = 128*~~

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- ~~*DirSwitchN = 255*~~
- ~~*AdapterTimeSyncInterval = 16*~~
- ~~*OffsetAvgConst = 4*~~
- ~~*DeltaAvgConst = 0*~~